

- Executive Summary
- The Privacy Impact Assessment (PIA) evaluates privacy and data protection risks related to the Application, based on provided workflow steps involving customer and order data processing.
- Project Overview and Purpose of Processing
- Processing involves reading customer and order CSV files and loading them into Delta tables for further data transformation, aggregation, and historical record-keeping.
- The primary objective is to process, join, aggregate, and manage customer and order data to enable analytics.
- Description of Personal Data Categories Processed
- Customer Data:
 - CustId
 - Name
 - EmailId
 - Region
- Order Data:
 - OrderId
 - ItemName
 - PricePerUnit
 - Qty
 - Date
- Data Flow Architecture (collection, storage, processing, sharing)
- Data is collected from CSV files from volumes.
- Data is loaded into Delta tables: customer, order, ordersummary, customeraggregatespend.
- Schema changes and aggregations occur during processing, with tables updated for SCD Type 2 logic.
- No explicit data sharing outside the processing environment is described.
- Lawful Basis for Processing
- Data Minimization Approach
 - Only specified fields are processed and stored.
 - Null and duplicate records are removed to maintain data integrity.
 - Aggregation and transformation steps are limited to requirements outlined.
- Consent Management Controls
- Third-Party / Vendor Data Sharing Implications

- Risk Identification:
- Confidentiality risks
- Potential exposure of customer EmailId, Name, and Region during processing or unauthorized access.
- Integrity risks
- Data corruption risk during transformation steps (e.g., wrong aggregations or missed updates).
- Re-identification risks
- Use of unique identifiers (CustId, EmailId) could allow re-identification.
- Access management risks
- Risk of improper permissions on Delta tables, unaccounted sharing of processing pipelines.
- Impact Assessment with Severity & Likelihood Scoring
- Mitigation Strategies & Security Controls (Encryption, DLP, IAM, Logging)
- Enforce access controls and role-based permissions on Delta tables.
- Implement logging of access and data transformations.
- Encrypt storage volumes and secure transfer channels.
- Regular review and validation of processing pipelines.
- Remove or mask PII in non-production environments.
- Residual Risk Evaluation
- Compliance Mapping (GDPR Articles, Local Regulations)
- Approval & Sign-off Matrix